

Journal Report

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Research Topic:

SafeSight. To create a physical system that can take in images from fisheye cameras, process it, and then using models to check for any dangerous situations. That would mean the system would be have blindspot detection, rear and forward collision warning, lane drift warning, and a screen that could display position of other cars relative to the main car and be the warning system.

Mid October Goal: (What will you have done by 10/15-ish)

By mid october, two classes, I want to be able to run the yolov8n model on the training data fully without errors and accurately, so then later I can use the bounding boxes and distance for the blindspot detection.

October Goal: (And by 10/30-ish)

By end of october, I still want to go for my goal with having the blindspot detection to be completed, but not the screen integrated with the blindspot detection yet. This is because I will be getting a new system to use instead of the Raspberry pi 3, and also because I did not expect it would take me this much time to get this point currently.

Daily Log (10/12/25-10/19/25)

Tuesday October 21

I was trying to get the radial undistortion working and it got to the point where it was ok, but I decided to stop using. This is because it was taking many times longer to undistort it with this different method than using openCV and this issue would only become bigger when testing on raspberry pi 5.

Thursday October 23

I was in a meeting with my mentor the whole class. I think it went pretty well as I talked a lot about my project, and she gave me a lot of feedback for presenting in the future. Most importantly, she pointed out an issue that I would have to sort out in the future, which is where should I place the screen as it can not be distracting but also has to be useful.

Timeline

Date	Goal	Met?
Today minus 2 weeks	Test the yolov8n model and its accuracy and overall speed	Done
This week minus 1 week	Get the model fully working	Done
This week	Blindspot detection	Working on it
Today plus 1 week	Finish Blindspot detection (categorizing the possibility for merging)	TBD
Today plus 2 weeks	Test on Raspberry pi	TBD

Reflection

I am little behind as I was planning on getting my blindspot detection started on thursday but that was interrupted. I do not think it will matter that much because I already spaced out plenty time just in case I have bugs or something interrupts me.